Optimizing Thermal Performance in 2.5D Systems Using Embedded Isolators

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***Abstract*—This paper investigates thermal management in tightly integrated heterogeneous chiplet systems, focusing on a novel approach using embedded thermal isolators. In many 2.5D systems, such as modern enterprise GPUs, thermally sensitive chiplets like High Bandwidth Memory (HBM) are thermally coupled to high-power compute chiplets, leading to performance degradation. We propose and evaluate the use of thermal isolators embedded within the heat spreader to effectively thermally decouple chiplets. Our thermal simulations of a water-cooled 2.5D integrated GPU system indicate that conventional approaches like thermally-aware floorplanning are less effective due to the dominant heat transfer through the heat spreader. In contrast, our proposed thermal isolators can significantly increase thermal isolation between chiplets (by up to 61%), or even reduce overall average peak chip temperature (by up to 22.5%). We develop a closed-loop workflow incorporating thermal results to quantify performance impacts of thermal-induced throttling, finding that in an example GPU+HBM system, the isolator approach can yield performance gains of up to 37% for memory-bound workloads. These findings open up new avenues for thermal management and thermal-system co-optimization in 2.5D heterogeneous integrated systems, potentially enabling more efficient and higher-performing chiplet-based architectures.**

***Index Terms*—2.5D Integration, Thermal-aware floorplanning, Thermal isolation, Chiplet integration, Thermal throttling**

I. Introduction

2.5D integration, where multiple silicon dies are integrated on a single interposer, has become a key technology to enable the latest high-performance systems-on-chip (SoC) and graphics processing units (GPUs) [1]–[3]. This integration approach allows heterogeneous dies, such as logic, memory, and sensors, to be tightly integrated in a single package, providing benefits in terms of performance, power efficiency, and form factor [4].

However, a key challenge with 2.5D integration is managing the thermal dissipation of the multiple high-power chiplets within the package. Chiplet-based designs are especially sensitive to thermal issues because of the close proximity of heterogeneous dies and the shared heatspreader. Specifically, the simultaneous integration of high-power components such as compute dies along with temperature-sensitive devices such as High Bandwidth Memory (HBM) stacks, photonic transceivers, image sensors, and RF devices can lead to unacceptable degradation in performance and reliability. For example, HBM chiplets must aggressively increase their refresh frequency as the temperature increases, up to 4x the nominal refresh frequency [5], which, as shown later, has a significant performance impact.

This paper investigates a novel approach to address these thermal challenges: the use of thermal isolators embedded within the heat spreader. This method aims to effectively decouple chiplets thermally, mitigating the performance degradation caused by thermal coupling.

Previous work has addressed the thermal challenges of 2.5D/3D integration primarily through thermal-driven floorplanning techniques [6]–[13]. These techniques show promise in reducing peak system temperatures; however, as this paper shows, they often necessitate larger interposer or package sizes, which can be impractical for large designs.

Various other thermal isolation strategies have also been explored. For example, thermal isolation air gaps within 3D stacks have been studied [14], [15], but this method is specific to 3D stacked chiplets and lacks generalizability to 2.5D systems.

Furthermore, recent work [16] has explored embedding microchannel cooling within the interposer to enhance secondary heat dissipation paths. This approach, while very effective, would require significant changes to interposer design, and does not explicitly isolate chiplets from each other, permitting the creation of hotspots.

[17]–[19] also contribute to this area by exploring differential cooling, as well as different interposer materials. Our work builds on this by suggesting that, while different interposer materials can provide some thermal isolation, their effectiveness is comparatively limited in densely packed GPU+HBM configurations.

Additionally, [20] addresses thermal isolation using dielectric coolants within a microfluidic cooling framework, focusing on specific configurations and materials that may not be widely applicable, especially for general purpose 2.5D designs.

The utilization of thermal isolation materials, such as aerogel, as a method of isolation has also been explored. [21] proposed a way to utilize aerogels to protect a heat-sensitive MEMS sensor from its processing electronics. In addition, [22] explored the use of aerogel as an isolation material within the interposer. Although promising, this approach neglects to model the significant thermal coupling caused by the thick heat spreader, which, as we argue, is the primary medium for thermal coupling in 2.5D systems.

Lastly, [24] investigated thermal management using high conductivity materials to create new thermal paths between chiplets. Although this method is informative, our findings emphasize the importance of directly addressing thermal coupling pathways, particularly within the heat spreader, to fully mitigate thermal issues.

Our work builds upon and extends these approaches by focusing on the insertion of thermal isolators embedded within the heat spreader, which we demonstrate to be highly effective in preventing heat transfer to temperature-sensitive chiplets. While thermal isolators within a heat spreader are commercially available [23], as far as we can ascertain, this is the first work that has proposed and quantified their integration in a 2.5D computing context. The key contributions of this work are as follows:

1) We demonstrate, through careful thermal simulation, that the majority of inter-chiplet heat transfer occurs through the high thermal conductivity heat spreader rather than the interposer. This finding explains the limited benefits of thermally-driven floorplanning approaches in 2.5D systems and the effectiveness of isolators.
2) We propose and evaluate the use of thermal isolators embedded within the heat spreader, showing that this technique is far more effective in preventing heat transfer to temperature-sensitive chiplets compared to conventional approaches.
3) We develop a closed-loop tool flow that links thermal simulation [27] with performance simulation of GPU [28], [29] and HBM [30]. Using this flow, we demonstrate that thermal coupling in GPU-HBM interposer systems can cause up to 38% performance degradation for some General Purpose GPU (GPGPU) workloads, which reduces to only 1%-3% with the insertion of thermal isolators.

The paper is organized as follows. Section II motivates the need for thermal isolation with an HBM example. Section III introduces our embedded isolator approach and compares it with conventional isolation techniques. Section IV quantifies the performance impact of thermal coupling and isolation for an example GPU-HBM system on GPGPU workloads. Finally, Section V concludes the paper.

II. Thermal Coupling Pitfalls: Motivating Thermal Isolators

Thermal coupling between chiplets is a critical concern for heterogeneously integrated systems, particularly in configurations with heat-sensitive High Bandwidth Memory (HBM) chiplets co-integrated with high-power CPU/GPU chiplets. This section examines the performance impact of thermal coupling on HBM, highlighting the need for effective thermal isolation strategies such as our proposed embedded isolator approach.

Many machine learning accelerators have a high power compute chiplet next to HBM chiplets on a silicon interposer. HBM consists of stacks of Dynamic Random Access Memory (DRAM), a capacitative memory technology. Capacitor leakage increases with temperature, reducing memory cell retention time [31]. Thus, the frequency of row refresh operations at high temperatures must increase, as noted in the JEDEC specifications for HBM2 and HBM3 [32], [5].

According to JEDEC, HBM refresh behavior consists of two manufacturer dependent trip point temperatures, beyond which refresh rates must be doubled. Previous works have reported that these trip temperatures lie between 75-95 degrees [33]–[36]. For the purposes of this analysis, it is assumed that the trip points are at 75 and 85 degrees C. This is shown in Table I.

TABLE I: Temperature-compensated refresh trip points

Refresh Rate	Temperature Trip Point (°C)
$1\times$ tREFI	< 75
$0.5\times$ tREFI	75
$0.25\times$ tREFI	85

We use DRAMSim3 [30] to quantify the performance impact of HBM throttling by varying the temperature (and the subsequent refresh rate) while executing maximum-intensity HBM workloads (optimistic scenario). Table II shows that the HBM bandwidth decreases by 27%, read latency increases by 70%, and refresh energy increases by 400%.

TABLE II: Normalized HBM performance at different refresh rates

tREFI Ratio	Refresh Energy	Read Latency	Bandwidth
1.00	1.000	1.000	1.000
0.50	2.000	1.238	0.912
0.25	4.004	1.714	0.732

This lower effective bandwidth is potentially very limiting, especially when executing memory-bound workloads, which are increasingly important and common in the ML field [38]. Section IV later shows the thermal coupling-driven slowdown caused in the application kernels for a GPU-HBM system that can exceed 30%. These significant performance impacts underscore the critical need for effective thermal isolation strategies. In the next section, we compare several thermal isolation approaches, with a focus on our proposed method of embedding thermal isolators within the heat spreader.

III. Comparative Thermal Analysis of Isolators and Conventional Isolation Approaches

We investigate the impact of various thermal management and isolation approaches, including:

- Thermal floorplanning: Optimizing chiplet placement to minimize thermal coupling [7]–[9].
- Naive thermal "spreading" of chiplets: Attempt to reduce thermal effects by spreading chiplets apart without any sophisticated thermal floorplanning algorithm. This aims to represent a more simplified version of thermal

floorplanning. It preserves the topological relationship between chiplets while reducing thermal coupling.

- Thermal isolators: An emerging approach in which thermal isolators are introduced in the heatspreader to enhance thermal isolation. [14], [15], [22].

Although we will examine all three approaches, our primary focus is on the thermal isolator strategy. We will demonstrate that this method offers superior thermal isolation in 2.5D systems, particularly in scenarios where traditional approaches like floorplanning and spreading are limited by practical constraints.

A. Methodology and Thermal Simulation Setup

To evaluate the effectiveness of various thermal management strategies, particularly our proposed thermal isolator approach, we performed detailed thermal simulations on two distinct systems: a GPU+HBM system inspired by the NVIDIA V100 GPU and a more complex heterogeneously integrated (HI) system based on the TAP2.5D design [8]. This section outlines our simulation methodology and setup for both systems.

Our first test model is a GPU+HBM system inspired by the NVIDIA V100 GPU.Each HBM chiplet stack is 4GB of memory. The specifications for this system are detailed in Table III.

TABLE III: V100-Inspired System Specifications

Chiplet	GPU	HBM
Width $[mm]$	32	7.56
Height $[mm]$	25.5	11.49
Power $[W]$	202	12
Number of Chiplets	1	4

The top-down configuration of this model is illustrated in Figure 1.

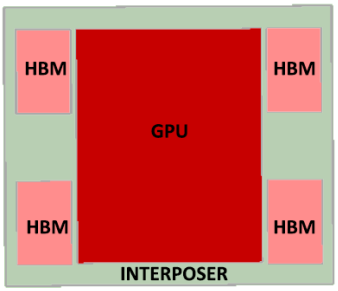


Fig. 1: V100-Inspired model setup. The central die is the GPU, surrounded by 4 HBM dies.

In our model:

- The compute chiplet is modeled as monolithic silicon.
- The HBM chiplet is modeled as an 8-tall DRAM stack with a $100\mu m$ buffer I/O die (2.4W), eight $50\mu m$ DRAM dies (1.2W each) and $20\mu m$ bonding layers between DRAM dies, aligning with configurations in [24]. This is illustrated in Figure 2.
- Total HBM energy consumption is set to $7pJ/bit$ [39].
- The chiplets are placed on a silicon interposer, which sits on an FR-4 package/PCB with modest backside cooling ($10W/mK$). This represents natural free air convection. [37].
- The PCB and heat spreader measure 266mm x 112mm, following standard GPU card form factors.
- A 0.1mm thick thermal interface material (TIM) layer ($100W/mK$) is placed on top of the dies, followed by a 1mm thick copper indentation connecting to the heat spreader.
- A thin 0.02mm bonding layer rests between the GPU/HBM chiplets and the interposer. The conductivity of this bonding layer is set to 10W/mK.

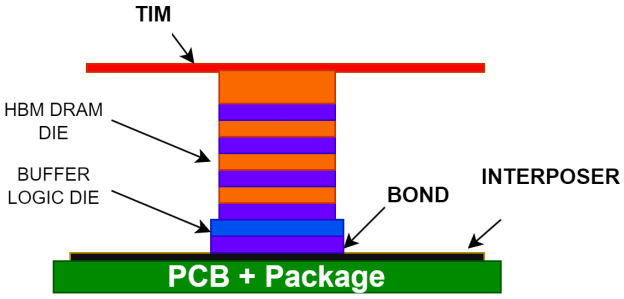


Fig. 2: Indicative HBM thermal model. Not to scale.

It is important to quantify the exact thermal rejection characteristics of the heat spreader, as it may affect the validity of further results and conclusions.

The heat spreader consists of a 15mm thick copper block with nine $8mm \times 4mm$ built-in cooling channels fed by a water pump. The average flow velocity is 0.008m/s, which was chosen empirically to adequately cool the GPU+HBM system. Using an empirical correlation from [45] to estimate the heat transfer coefficient (HTC) of the system for laminar flow, we get an estimated value of $h = 341.5$W/m^{2}K. The flow is laminar according to thermal simulator and CFD results (Reynolds number ≈ 5).

Our second test model is a more complex heterogeneously integrated system inspired by TAP2.5D [8]. The specifications for this system are detailed in Table IV, and a top-down view is visible in Figure 13a.

TABLE IV: Complex HI System Specifications [8]

Chiplet	CPU	GPU	HBM
Widths $[mm]$	12	18.2	7.75
Height $[mm]$	12	18.2	11.87
Power $[W]$	105	295	20
Number of Chiplets	1	2	3

For this system, we maintain the same overall cooling setup as the GPU+HBM system, but increase average water velocity to 0.5m/s across all 9 channels to handle the higher total power.

All thermal simulations were conducted using ANSYS Icepak [27]. We modeled various thermal management strategies for both systems:

- Baseline configuration
- Naive spreading of chiplets
- Thermal floorplanning (for the Complex HI system)
- Thermal isolators with various placements

B. Thermal Isolators: Concept and Implementation

The core concept of our thermal management strategy involves the introduction of thermal isolators within the heat spreader to effectively "split" the thermal pathways between chiplets. This approach aims to minimize thermal coupling between high-power compute dies and temperature-sensitive components such as HBM. To understand the effectiveness of this approach, it is crucial to first examine the heat transfer pathways in a typical 2.5D system. In a typical 2.5D integrated system, there are multiple heat transfer pathways, as illustrated in Figure 3. The primary pathways are:

1) Conduction through the interposer and PCB/package
2) Minimal natural convection in the gap between chiplets
3) Conduction through the thermal interface material (TIM) and copper heat spreader

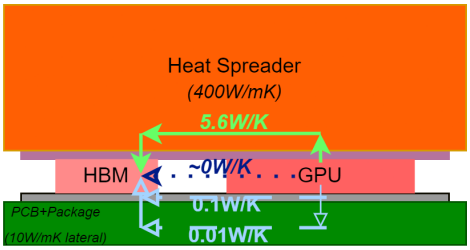


Fig. 3: Heat pathways in a typical thermal setup. Each arrow represents a thermal pathway, and the number indicates the thermal conductivity multiplied by the cross-section (thickness) of the pathway.

Among these, the conduction path through the TIM and heat spreader is the most significant due to the high thermal conductivity of copper, allowing efficient lateral heat transfer between chiplets. When large thermal gradients exist, like those on heterogeneous 2.5D chips, this is counterproductive, as it also increases the thermal coupling between them. Conventional thermal management techniques, such as floorplanning and spreading, offer limited isolation benefits since increasing chiplet spacing does not significantly affect the dominant heat-spreader-based pathway.

To support this conclusion we have conducted simulations using the GPU+HBM system to identify the concrete impact of changing TIM and interposer materials. Our analysis suggests that the thermal conductivity of the thin TIM layer has a minimal impact on overall results. Figure 4 shows the $\Delta Temperature$ between GPU and HBM for TIM conductivities ranging from 10 W/mK to 300 W/mK. A conventional high performance TIM is around 20 W/mK [40]. Across this range, the difference in $\Delta Temperature$ never exceeds 1.25 °C, indicating that changing TIM quality alone is insufficient to achieve meaningful thermal isolation. Throughout our study, we use a TIM conductivity of 100 W/mK as a representative value of high-end TIM.

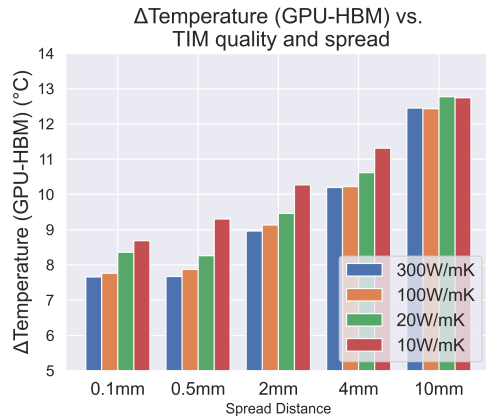


Fig. 4: $\Delta Temperature$ vs. TIM conductivity. The difference between GPU and HBM temperatures remains nearly constant across a wide range of TIM conductivities, underscoring the minimal influence of TIM quality on thermal isolation.

We further examine the effect of interposer material on thermal coupling. This result may be dependent on the thermal conductivity of the bonding layer between the interposer and the GPU/HBM chiplets. For the base case of a 10 W/mK bonding layer, Figure 5 shows that changing the interposer from Silicon ($148\,W/mK$) to Glass ($1.5\,W/mK$) or FR-4 ($0.3\,W/mK$) reduces the peak HBM temperature by only up to 2.5 °C and has a small effect on GPU temperature. However the thermal conductivity of the bonding layer may be higher in practice, as microbumps provide excellent thermal conduction pathways to the interposer. This may alter our conclusions.

Figure 6 showcases that even if the bond conductivity is increased from 10 W/mK to $401W/mK$ (as high as pure copper), the $\Delta Temperature$ between the GPU and HBM does not change significantly regardless of interposer material. This demonstrates that even in the pessimistic high conductivity bond case, merely choosing a less conductive interposer material does not provide substantial isolation benefits, as the core heat conduction path through the heat spreader continues to dominate.

Furthermore, this conclusion is not altered by the bonding layer's thickness, which can range from our assumed $20\mu m$ up to $130\mu m$ in practice [25], [26]. Thicker bonding layers may reduce lateral thermal resistance through the infill/underfill (which has much poorer thermal conductivity than silicon) albeit at the cost of increased thermal resistance to the interposer. Our analysis of the extreme $401W/mK$ case (which represents a far lower thermal resistance path than a thicker standard bond) already demonstrated that this pathway does not meaningfully contribute to inter-chiplet coupling.

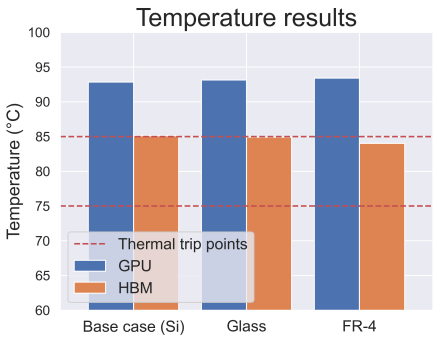


Fig. 5: Interposer material comparison. Switching from Silicon to lower-conductivity materials (e.g., Glass, FR-4) results in only minimal temperature reduction, illustrating the limited role of interposer material choice in enhancing thermal isolation.

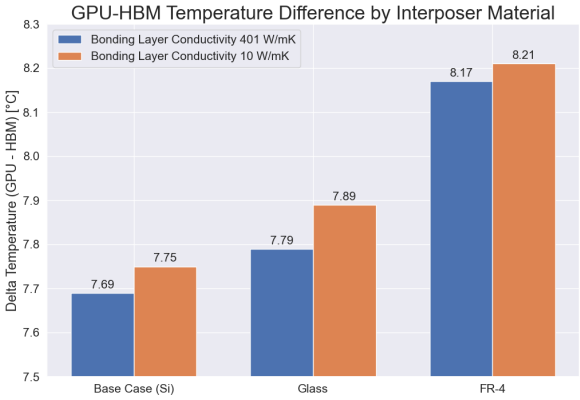


Fig. 6: Interposer material comparison for two different bond conductivities. Even in a very high bonding conductivity scenario (401 W/mK), interposer material changes do not effectively isolate the chiplets.

These observations collectively highlight that neither TIM conductivity tweaks, increased chiplet spacing, nor less conductive interposer materials substantially curb inter-chiplet heat flow. Since the thick copper heat spreader dominates lateral heat transfer, none of these conventional strategies achieve meaningful thermal isolation.

Our proposed solution leverages this insight. Instead of relying on spacing or material changes, we introduce low-conductivity thermal isolators within the heat spreader to effectively "split" it into distinct thermal zones. As shown in Figure 7, inserting isolators significantly reduces the lateral heat flow between chiplets. This approach isolates temperature-sensitive components much more effectively than conventional methods, enabling finer-grained thermal management in tightly integrated 2.5D systems.

In our study, we explored various thermal isolation material approaches, including 0.1mm-thick aerogel-based isolators, 0.1mm-thick air gaps, and 1mm-thick isolators with the conductivity of FR-4. As shown in Fig 8, all of these isolator techniques serve the same function of severing virtually all thermal

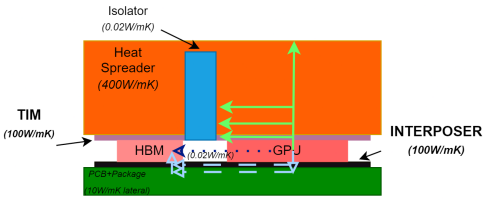


Fig. 7: Setup with isolators. The inserted isolators create separate thermal zones, diminishing lateral coupling in the heat spreader.

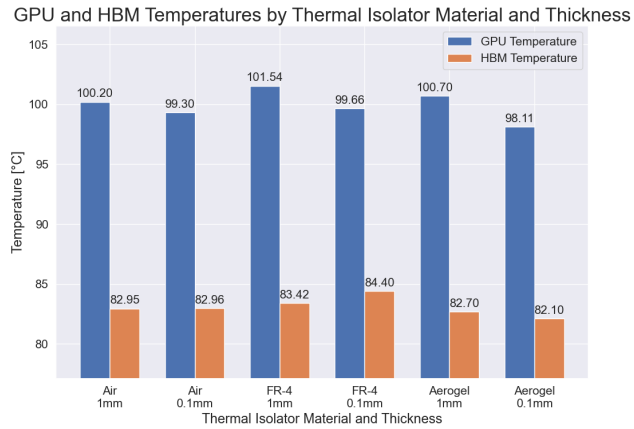


Fig. 8: Comparing different isolator designs by their ability to isolate GPU/HBM chiplets in an example scenario. Thicker isolators take up more space within the heat spreader, reducing it's effectiveness, but are helpful to ensure isolation for more conducting materials like FR-4. However, all isolators tested behave similarly thermally (differences $\leq 2°\mathrm{C}$)

coupling across them, with very similar thermal results. The results presented in this paper assume the 0.1mm-thick aerogel isolators.

This approach is similar to that proposed in [22], but instead of limiting the isolating material to the interposer, we place it in the copper heat spreader. For systems with larger inter-chiplet distances ($> 1mm$), this could be implemented by literally splitting the heat sink into multiple separate heat spreaders. The integration of isolators within the heat spreader introduces potential challenges in terms of structural stability and manufacturability. To mitigate these concerns, the designs incorporate copper overhangs around the isolators, ensuring mechanical stability without compromising thermal isolation. Their placement is illustrated in Figure 9 and Figure 10 for the example V100-style system.

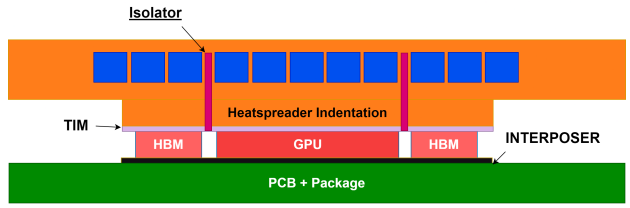


Fig. 9: Isolator setup. Not to scale, only indicative of isolator placement.

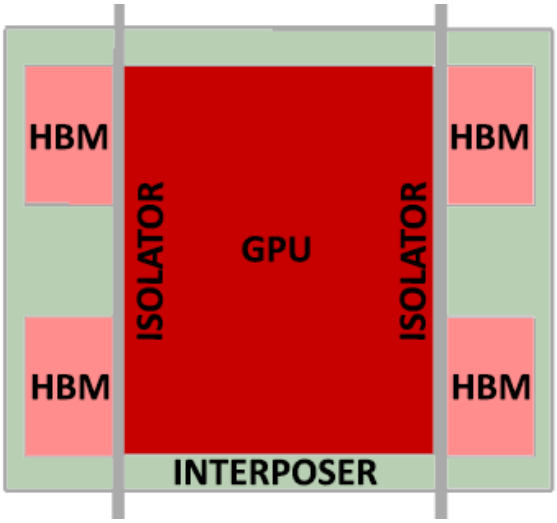


Fig. 10: Top-down isolator setup. Not to scale, only indicative of isolator placement.

The placement of thermal isolators is crucial and depends on the specific system configuration and thermal management goals. We identified two primary objectives for isolator placement:

1) Reducing overall peak system temperature, especially around high-power dies
2) Protecting temperature-sensitive components (e.g., HBM) from high-power compute dies

Based on these objectives, we developed different placement strategies for our two test systems:

- **GPU+HBM System:** In this simpler configuration, our primary goal was to thermally isolate the HBM from the high-power GPU. Therefore, we placed isolators between the GPU and HBM chiplets, effectively creating separate thermal zones for each component.
- **Complex HI System:** This more intricate system allowed for multiple isolator placement strategies:
  - Placement between high-power compute dies (CPU and GPUs) to manage peak temperatures and thermal gradients across the entire system
  - Placement between HBM and the high-power compute dies to protect the temperature-sensitive HBM

By strategically placing these isolators, we aim to create distinct thermal zones within the heat spreader, effectively minimizing heat coupling. The following sections will present the results of our thermal simulations, demonstrating the effectiveness of this approach in both the GPU+HBM and Complex HI systems compared to conventional approaches.

C. Thermal Results for GPU+HBM System

Figure 11 shows that while distancing the chiplets from the base case of 0.5mm to up to 4mm, the temperature difference is minimal and never exceeds 3 degrees C for the GPU die or the HBM chiplets. Meaningful differences are only prevalent at 10mm of separation, which is highly impractical due to the large package and interposer size increase it necessitates. As shown in Figure 11, to physically fit the chiplets while spreading, the underlying interposer had to increase in size by 11%, 22% and 55% for 2mm, 4mm and 10mm separation, respectively. This indicates that spreading is a poor strategy in thermal isolation, which corresponds to our assumptions about the importance of the top lateral thermal pathway. Thermal isolation through separation alone would require unreasonably large distancing between chiplets, which would result in manufacturing/packaging issues and cost increases (due to larger interposer size), as well as increased communication link power and decreased performance (due to longer communication links in the interposer).

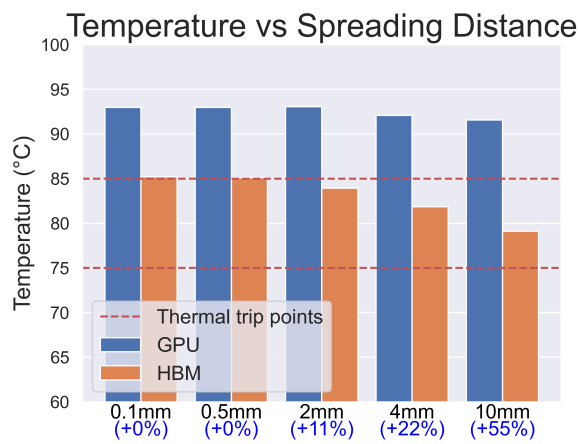


Fig. 11: Temperature vs Spreading Distance. Percentage below spread distance indicates additional interposer/package area required.

These results indicate that spreading is a poor strategy for thermal isolation in densely packed 2.5D systems. Achieving meaningful isolation through separation alone demands unfeasibly large distances, leading to manufacturing/packaging issues, higher costs, and increased communication power.

In contrast to the limited benefits of spreading, our proposed thermal isolator approach shows significant improvements in thermal management. Figure 12 presents the thermal results for the V100-inspired GPU+HBM system with and without thermal isolators.

The proposed isolators improve isolation and effectively split the heatspreader into distinct thermal zones between the chiplets. From a baseline temperature of 85.1 degrees, surpassing both HBM "trip points", incorporating an isolator within the heat spreader effectively reduces the HBM temperature to 71.76 degrees. However, this increases GPU temperature significantly, which would necessitate throttling the GPU. We explore the performance implications of this in Section IV.

These results demonstrate the superior effectiveness of thermal isolators in managing heat distribution in densely packed 2.5D systems like our GPU+HBM model. By addressing the primary heat transfer pathway through the heat spreader,

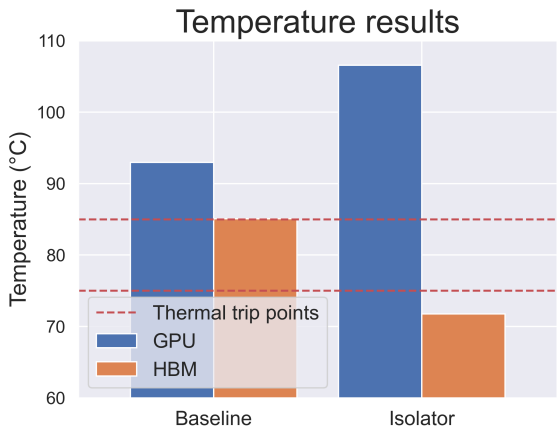


Fig. 12: Baseline vs thermal isolator results

isolators provide a more practical and efficient solution to thermal management challenges in these systems.

D. Thermal Results for Complex Heterogeneously Integrated System

To validate the generality of our findings, we extended our thermal management study to a more complex heterogeneously integrated (HI) system inspired by TAP2.5D [8]. For this complex HI system, we evaluated several thermal management strategies, as shown in 13.

Figure 14 presents the results of the thermal simulation for the different management strategies:

Our analysis of these results reveals several key insights:

- **Baseline and Conventional Approaches:**
  - Both thermal floorplanning and naive spreading have very modest isolation effects.
  - Both techniques lower peak and HBM temperatures, but require a 38.4% larger interposer area [8], which is impractical for most designs.
- **Thermal Isolator Strategies:**
  - *Isolators around HBM* (Figure 13b): Effectively shields HBMs from compute die heat, reducing HBM temperature below the first trip point without increasing interposer area.
  - *Isolators around compute dies* (Figure 13c): Shows modest decreases in both HBM and compute temperatures without increasing interposer area. The isolators effectively eliminate the central hotspot by isolating the high power dies from each other.
  - *Isolators around compute dies with 18% larger interposer* (Figure 13d): Significantly lowers both HBM and peak compute temperatures with only a modest increase in interposer size (18% vs. 38.4% for other methods).

Comparing the different approaches, we observe that:

- Isolator strategies can be tailored to specific thermal management goals (e.g., reducing HBM temperature or overall peak temperature).
- Thermal isolators provide superior thermal isolation, especially for temperature-sensitive components like HBM. In this case, the ΔT between HBM and the compute dies increased by up to 61% from Baseline to the T.Isolator (HBM) case.
- Isolators can achieve thermal improvements, even reducing peak overall compute temperature by 3.5%, with no increase in interposer size. This is achieved by preventing the creation of thermal hotspots and more effectively amortizing the heat rejection across the entire heat spreader.
- When combined with a modest increase in interposer size, Isolators provide even better thermal improvements. In the T. Isolator (PT,LI) case, peak overall compute temperature was decreased by 22.5%.

These results underscore the effectiveness and flexibility of thermal isolators in managing heat distribution in complex 2.5D integrated systems. By creating distinct thermal zones, isolators offer a more targeted and efficient approach to thermal management, outperforming conventional strategies such as spreading and thermal floorplanning.

IV. Improved System Performance by Thermal Isolators: A GPU+HBM Example

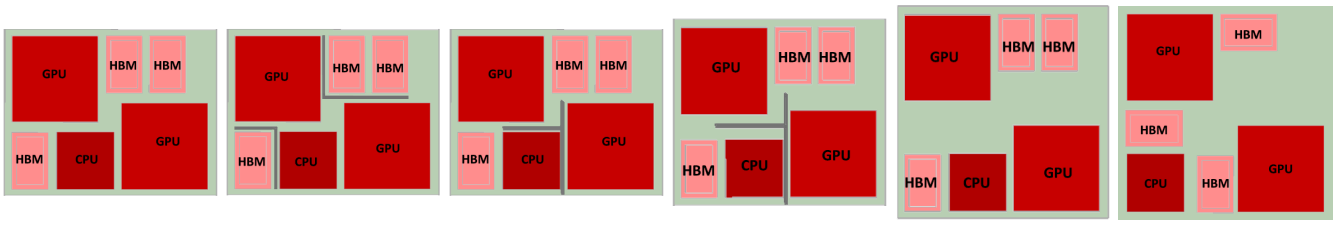
In Section II, we demonstrate how thermal heating of HBM chiplets can cause them to exceed their thermal design trip points, severely affecting their performance. Section III showcased how thermal isolators can shield HBM and other heat-sensitive chiplets from power-hungry compute dies in a 2.5D integration environment, mitigating this effect.

We now evaluate the potential impacts of improved thermal isolation during the execution of AI workloads, illustrating the efficacy of the proposed thermal isolator-in-heatspreader design. Our focus is on algorithms that are both computationally intensive (high power) and memory-intensive (high bandwidth), as these workloads are particularly susceptible to thermal coupling effects in 2.5D integrated systems.

A. Methodology and Setup

The chosen platform for this analysis is a GPU-HBM system on a silicon interposer, similar to the Nvidia Titan V GPU. Notably, the Nvidia Titan V employs the identical package architecture and maintains the same Thermal Design Power (TDP) as the GPU+HBM system highlighted in previous results. This allows us to apply the thermal data presented in Section III to our selected system.

To quantify the performance impact of thermal-driven throttling of compute and memory on real workloads, we utilized a workflow that incorporates Icepak [27], DRAMSim3 [30],



(a) Baseline (b) HBM Isolators (c) Comp. Isolators (d) Comp. Isolators + Larger Interposer (e) Naive Spreading (f) Thermal Floorplanning

Fig. 13: Comparison of different Complex HI model setups highlighting various configurations and their thermal management strategies. (a) Baseline configuration. (b) Thermal isolators around HBM, shielding HBM chiplets from compute die heat. (c) Thermal isolators around compute dies, reducing overall system temperatures without increasing interposer area. (d) 18% larger interposer with thermal isolators around compute dies, enhancing thermal management. (e) Naive spreading of chiplets. (f) Thermal floorplanning (as proposed in TAP2.5D [8]).

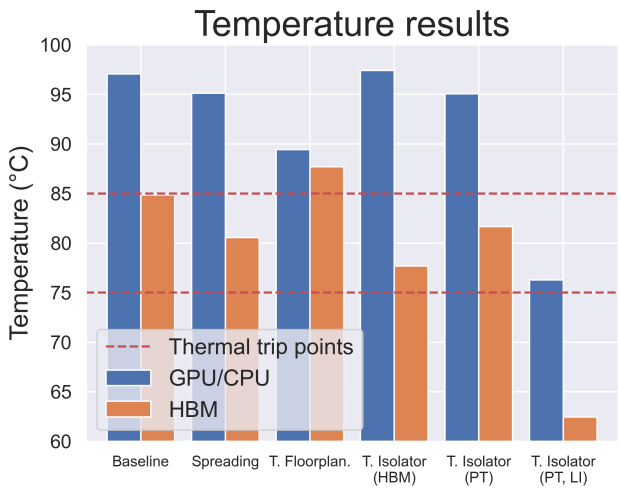


Fig. 14: Complex HI model thermal management results. Abbreviations: Spreading → Naive Spreading with 38% increased interposer size [Fig 13e]. T. Floorplan → Thermal Floorplanning with 38% increased interposer size [Fig 13f]. T. Isolator (HBM) → Thermal Isolator placed around HBM [Fig 13b]. T. Isolator (PT) → Thermal Isolator placed around compute dies for Peak Temperature reduction [Fig 13c]. T. Isolator (PT,LI) → Thermal Isolator placed around compute dies, with a 18% Larger Interposer [Fig 13d].

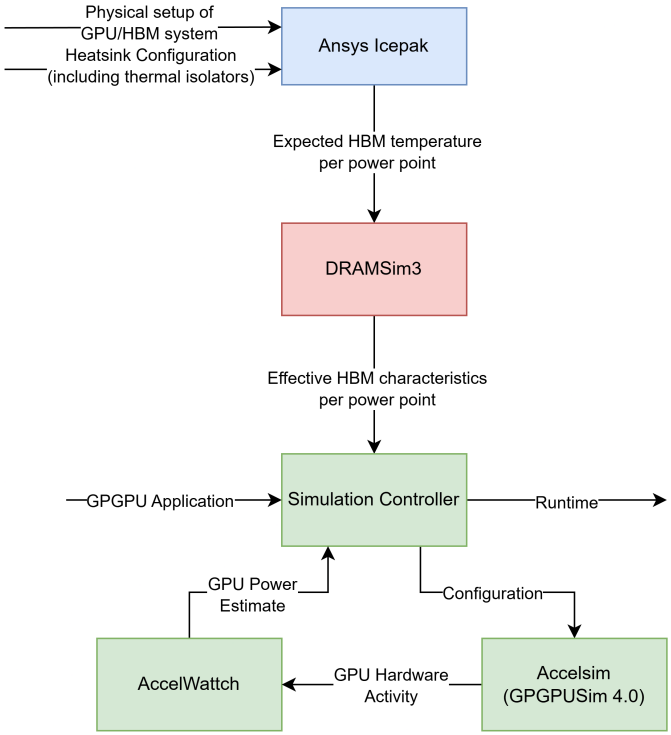


Fig. 15: Performance workflow

Accelsim [28] and Accelwattch [29] together. An illustration of this workflow is shown in Fig. 15.

First, we utilize a thermal simulator, such as Ansys Icepak, to create a table of GPU power vs. steady-statetemperature for a given GPU system and heatspreader configuration that we wish to simulate. Two example tables (baseline / isolators) for the Nvidia TitanV GPU are shown in Tables V and VI.

TABLE V: Baseline Power-Temperature table

Power [W]	GPU [°C]	HBM [°C]
202	92.97	85.10
168	86.14	80.68
134	79.21	76.57

Leveraging the data from these example Tables, as well as Tables I and II as detailed in Section II, we established

TABLE VI: Thermal Isolator Power-Temperature table

Power [W]	GPU [°C]	HBM [°C]
202	103.34	74.39
168	94.07	72.15
134	84.76	69.91

a correlation between power consumption and performance metrics. Specifically, the following observations are made, assuming a maximum GPU temperature of 95 degrees:

- In the baseline configuration, HBM thermal throttling occurs at all examined GPU power levels (134W to 202W), while GPU temperatures remain below throttling thresholds.
- With thermal isolators, HBM doesn't throttle across the

examined power range. However, GPU throttling occurs when power exceeds 168W due to higher GPU temperatures.

To further our analysis, we initiated simulations of a specified workload using Accelsim [28]. These simulations generated a heatmap of multiple scenarios at different throttle points for the GPU (due to the decrease in clock frequency caused by overheating) and the HBM (due to increased refresh rates when trip-points are exceeded).

Subsequently, employing Accelwattch [29] - a power estimation tool integrated within Accelsim - we extracted the estimated power consumption of the workload in question. This data allowed us to implement selective throttling within Accelsim based on the observed power consumption.

Due to the relatively rudimentary HBM model in Accelsim, direct modifications to the number of refresh operations were not feasible. Consequently, we approximated the increase in refresh rates by simulating reduced bandwidth (achieved by decreasing the effective HBM clock speed) and increased read latency. We also simulated GPU throttling by simply reducing GPU compute clock frequency following quadratic frequency-power scaling.

We then compared execution times between the no-isolation case and the isolator case while taking HBM and GPU throttling into account.

B. Results and Sensitivity Analysis

We benchmarked Polybench-conv2d [41], a convolutional filter application, and DeepBench inference [42], a general-purpose GEMM workload. Both are examples of high-power yet memory-bound algorithms. It must be stressed that only workloads that meet these two characteristics will optimally benefit from the insertion of isolators. Low-power workloads will not throttle the memory or GPU, and compute-bound algorithms can afford to ignore memory throttling and will instead suffer due to increased GPU throttling. To showcase this, we also list Polybench GEMM [41], which is a compute-bound GEMM workload. However, we want to stress that many modern ML workloads meet these criteria [46].

It is also important to note that the performance results are highly sensitive to our assumptions. Up to this point, we have used DRAMSim3 results that suggest large bandwidth reductions due to increased refresh rates. However, this may be an optimistic scenario, as DRAMSim3 assumes maximum memory traffic. Additionally, we have assumed relatively low HBM trip points (with the second trip point at 85°C), which may not hold for all HBM implementations.

To account for this variability, we presented a sensitivity analysis in Table VII. This table shows the performance implications for the benchmarks under different assumptions:

1) Large bandwidth reduction (as per DRAMSim3) vs. halved bandwidth reduction [1.0x BW vs 0.5x BW]
2) Low trip points (second trip at 85°C) vs. high trip points (second trip at 95°C) [Low Trip vs High Trip]

TABLE VII: Sensitivity analysis of performance improvements for Polybench CONV2D, Deepbench GEMM (D.GEMM), and Polybench GEMM (P.GEMM)

	Low Trip		High Trip	
Benchmark	1.0x BW	0.5x BW	1.0x BW	0.5x BW
`Conv2D 2048x2048`	29%	9%	4%	0%
`Conv2D 4096x4096`	37%	15%	8%	5%
`Conv2D 8192x8192`	33%	14%	7%	4%
`D.GEMM 1760x7133x1760`	7%	2%	7%	2%
`P.GEMM 1024x1024x1024`	-11%	-11%	-11%	-11%

This sensitivity analysis reveals that the performance benefits of thermal isolators can vary significantly based on our assumptions about HBM behavior under thermal stress. In the most optimistic scenario (large bandwidth reduction, low trip points), we see performance improvements of up to 37% for the 4096x4096 Conv2D benchmark. However, in more conservative scenarios, the improvements are more modest, showcasing up to only 5% speedup. Finally, in a scenario like Polybench GEMM, which is not memory bound at all, isolators lower performance and are not a good fit.

These results underscore the importance of thermal management in 2.5D integrated systems, particularly for high-power, high-bandwidth workloads. Although exact performance gains may vary based on specific HBM characteristics and thermal conditions, our thermal isolator approach provides meaningful improvements in a range of scenarios.

V. Discussion and Conclusions

Our results clearly demonstrate that thermally aware chiplet floorplanning is very limited in its effectiveness and requires large ($> 4mm$) separation between hot chiplets (e.g., GPU) and temperature-sensitive chiplets (e.g., HBM). This makes thermally-aware floorplanning impractical in most cases due to a communication performance penalty and increased interposer/package area. We show that the majority of thermal coupling occurs within the high thermal conductivity thick heatspreader rather than within the very low thermal conductivity thin interposer. Because of this, changing physical characteristics such as interposer materials or bonding conductivity/thickness does not have an impact in reducing overall thermal coupling.

We instead propose and analyze the insertion of thermal isolators within the heat spreader as a viable thermal isolation approach. The isolators studied consist of 3 different materials (Air, FR-4 and Aerogel), with very similar thermal performance characteristics. Notably, all 3 materials seem to be durable candidates as they tolerate thermal cycling well with minimal loss of thermal conductivity. Air-gaps do not degrade thermally, aerogel thermal conductivity only increases by $\approx 10\%$ even after aging [43], and FR-4 can withstand much higher thermal cycling temperatures than the ones in a 2.5D system [44].

We show that the targeted insertion of thermal isolators within a complex HI system can reduce peak temperature by 3.5% without any change in interposer area or 22.5% with an 18% increase in interposer area, while thermal floorplanning reduces peak temperature by only 9.98% with 38% interposer area overhead. We also showed that thermal isolator insertion in a GPU+HBM system can improve the performance of memory-bound machine learning workload kernels by as much as 37%.

While CPUs and GPUs are often the primary focus of conventional thermal management strategies due to their high power densities, this work demonstrates the critical importance of addressing thermal coupling in temperature-sensitive components like HBM. The increasing reliance on HBM for memory-bound and bandwidth-sensitive workloads, particularly in AI/ML and HPC systems (e.g., NVIDIA H100, AMD MI300), underscores the need for solutions that ensure consistent memory performance under thermal stress. The proposed isolator approach is not intended as a standalone solution but as a complementary technique that can be integrated alongside other thermal management methods to achieve balanced performance in heterogeneous 2.5D systems.

As such, this work opens up new avenues for thermal management and thermal-system co-optimization for 2.5D HI systems. The following are some examples of future work directions.

- Isolator placement in complex systems is challenging due to varying power consumption and temperature sensitivity of chiplets. In high power, thermally challenging systems, isolator placement and thermal floorplanning/spreading of chiplets may need to be co-optimized, even if this comes with some area overhead.
- This work has focused on microchannel liquid cooling setups in a 2.5D context. Physical thermal isolation in other cooling approaches such as air cooling, or different system architectures such as 3D need to be studied.
- This work has not considered the impacts of the larger temperature gradients that thermal isolators impose on the interposer and the resulting thermal expansion mismatches. This needs to be studied to ensure reliability and manufacturability.
- While this work considered a variety of thermal isolator materials during thermal simulations (Aerogel, FR-4, Air), it remains to be determined which material is most manufacturable and practical.

In conclusion, this paper highlights the critical importance of thermal-system co-optimization in tightly integrated 2.5D heterogeneous chiplet systems. We have demonstrated that the insertion of explicit thermal isolators to "split" heatspreaders offers a superior solution to thermal management challenges in these systems, outperforming conventional approaches like thermally-aware floorplanning. This novel approach paves the way for more efficient and higher-performing heterogeneous 2.5D integrated systems.

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